TEXORA · LECTURE SHEET

# Human Color Vision

*Three cone types, opponent channels and a brain that adapts — why the eye is a superb comparator but a poor measuring instrument, and why observers differ.*

WP-4103 · COLOR PHYSICS · TOPIC 3 · SPECIALIZATION — WET PROCESSING

## 1. The eye as an optical instrument

Light enters through the **cornea** and the adjustable **lens**, which focus an inverted image onto the **retina** at the back of the eye; the **iris** controls the pupil diameter and hence the light level. The retina converts the optical image into neural signals. Two features dominate colour work: the tiny central **fovea**, packed with cones, gives sharp, colour-rich vision over only about 2° of visual angle, while the wider retina trades colour detail for sensitivity. Signals leave via the **optic nerve** — creating a blind spot where it exits — to the visual cortex, where colour is finally constructed.

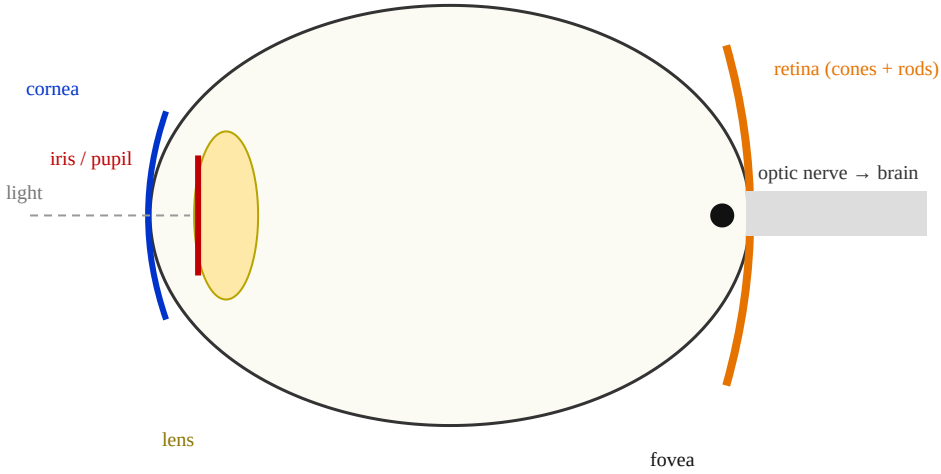



Figure 1 — Cross-section of the human eye (schematic): cornea, iris, lens and the retina with its cone-rich fovea.

## 2. Rods and cones — two receptor systems

The retina holds two photoreceptor types, about 120 million **rods** and 6–7 million **cones**:

| FEATURE | RODS | CONES |
|---|---|---|
| Number / location | ≈ 120 million, mostly peripheral retina | ≈ 6–7 million, concentrated at the fovea |
| Light level | Dim light (scotopic vision) | Bright light (photopic vision) |

| | | |
|---|---|---|
| Colour | None — shades of grey only | Full colour |
| Detail | Poor acuity, high sensitivity | High acuity |
| Types | One (pigment peak ≈ 498 nm) | Three: L, M and S cones (Section 3) |

Practical consequence: shades must never be judged in dim light, where rods take over and hue information disappears — one reason light cabinets specify a minimum illumination level.

## 3. Trichromatic theory

The **trichromatic theory** (Young–Helmholtz) states that colour vision rests on three cone types with overlapping spectral sensitivities peaking in the long-, middle- and short-wavelength regions:

- **L cones** — peak sensitivity near 564 nm ("red-sensitive", though the peak is in the yellow-green).
- **M cones** — peak near 534 nm (green region).
- **S cones** — peak near 420 nm (blue region); only about 2% of cones, absent from the very centre of the fovea.

Every colour produces a unique ratio of L : M : S responses, and that triple *is* the colour signal — which is why three primaries suffice for additive mixing (Topic 1) and why the CIE system is built on three tristimulus values (Topic 2). Two physically different spectra giving the same L : M : S ratio look identical: the physiological root of **metamerism** (Section 7).

## 4. Opponent-process theory

Trichromacy describes the retina's input stage, but perception itself is organised in opposition (Hering). After the cone stage, signals are recoded into three **opponent channels**:

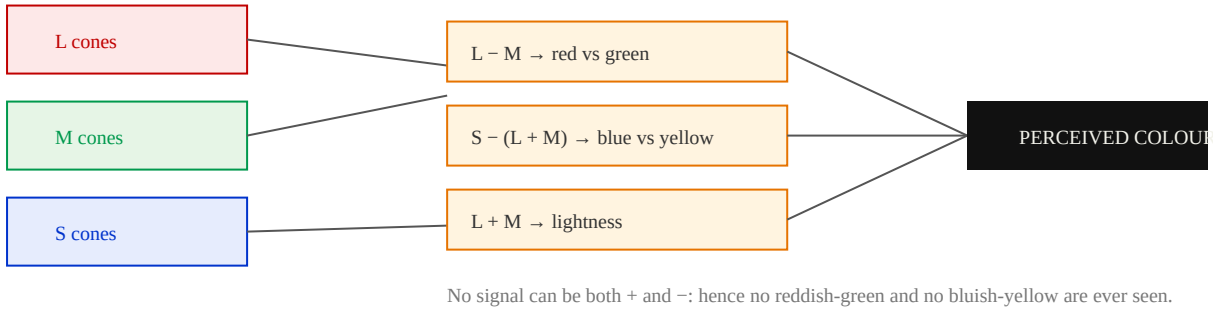



Figure 2 — Two-stage vision: three cone inputs (trichromatic) recoded into opponent channels — the model mirrored by CIELAB's a* and b* axes.

Opponent coding explains everyday facts trichromacy cannot: there is no reddish-green or bluish-yellow; staring at red fatigues the red side of the channel and produces a green **afterimage**; and

CIELAB's a* (red–green) and b* (yellow–blue) axes are deliberately built on this physiology. Modern understanding combines both theories in a **two-stage model**: trichromatic capture, opponent recoding.

## 5. Spectral sensitivity — V(λ), photopic and scotopic vision

The eye is not equally sensitive to all wavelengths. The CIE **luminous efficiency function V(λ)** peaks at **555 nm** (yellow-green) for daylight-adapted, cone-based **photopic** vision and falls to near zero at 380 and 780 nm — the formal definition of the visible limits. In dim light the rod system takes over (**scotopic** vision), the peak shifts to about **507 nm** and colour vanishes (the **Purkinje shift**: reds darken relative to blues at dusk, familiar when a red fabric looks almost black in low light while a blue one still reads). Between the two lies transitional *mesopic* vision. V(λ) is also embedded in colorimetry: the CIE $\bar{y}(\lambda)$ function is V(λ), which is why Y measures luminance.

## 6. Colour-vision deficiencies

About 8% of men and 0.5% of women inherit a cone deficiency: one pigment is altered (**anomalous trichromacy**, e.g. deuteranomaly — the commonest) or absent (**dichromacy**):

| DEFICIENCY | MISSING/ALTERED CONE | EFFECT |
|---|---|---|
| **Protanopia** | L cones absent | No red perception; reds appear dark, red–green confusion |
| **Deuteranopia** | M cones absent | Red–green confusion with normal brightness of red |
| **Tritanopia** | S cones absent (very rare) | Blue–yellow confusion |
| Achromatopsia | No functioning cones (extremely rare) | Total colour blindness, grey world, poor acuity |

Industrial consequence: anyone employed as a shade passer must be screened — the **Ishihara plates** for red–green defects and the **Farnsworth–Munsell 100 Hue test**, which also grades general colour-discrimination ability by how accurately chips are ordered. An undetected deficiency in a pass/fail role converts directly into customer claims.

## 7. Adaptation, contrast and metamerism

**Chromatic adaptation.** The visual system rescales cone gains so that a white object still looks white under tungsten or daylight — the reason standard illuminants can all look "white" in a cabinet. Adaptation is helpful but incomplete, and it needs time: assessors should adapt to each cabinet illuminant before judging, and never judge immediately after walking in from bright sun.

**Simultaneous contrast.** A colour is judged against its surround: the same grey fabric looks lighter on a black background and darker on a white one, and a neutral sample on a red surround takes a greenish cast (opponent induction). Assessment therefore uses a neutral grey surround and cabinet interior, samples in direct contact, and a fixed viewing geometry.

**Metamerism as a visual phenomenon.** Two samples with different reflectance curves can produce identical L : M : S responses — and therefore match — under one illuminant, then diverge when the illuminant changes the balance of the stimulus (**illuminant metamerism**: the classic D65-pass, TL84-fail). Observer metamerism arises because individual cone sensitivities differ slightly, especially between the 2° and 10° fields. Metameric matches are unavoidable when different dye sets are used; they are managed, not eliminated, by matching under multiple illuminants and choosing dyes close to those of the standard.

## 8. Observer variability and the standard observers

Because no two observers are identical, the CIE replaced people with tables. The **1931 standard observer (2°)** gives colour-matching functions $\bar{x}$, $\bar{y}$, $\bar{z}$ averaged from experiments using a 2° field — appropriate for small samples viewed at a distance. The **1964 supplementary standard observer (10°)** uses a 10° field, better matching how fabric panels are actually viewed, and is the observer specified for textile measurement: **D65 / 10°** is the industry default quoted in Topic 2. Individual variation around these averages — lens yellowing with age, macular pigment differences, deficiency — is why instrumental measurement, not a single trusted eye, arbitrates disputes, while trained, screened observers remain the final check because instruments do not judge texture and appearance.

## 9. Glossary — key terms

- **Retina / fovea:** light-sensitive layer of the eye; its cone-dense centre gives detailed colour vision.
- **Rods / cones:** receptors for dim-light grey vision and bright-light colour vision respectively.
- **Trichromatic theory:** colour vision from three cone types (L, M, S).
- **Opponent-process theory:** recoding into red–green, blue–yellow and lightness channels.
- **V(λ):** luminous efficiency function, photopic peak at 555 nm.
- **Purkinje shift:** peak sensitivity moves to ≈ 507 nm and colour is lost in scotopic vision.
- **Metamerism:** samples matching under one condition (illuminant/observer) but not another.
- **Standard observer:** CIE tabulated average observer, 2° (1931) or 10° (1964).

## 10. Key points

- Vision is two-stage: trichromatic capture by L, M, S cones, then opponent recoding — the design CIELAB copies.
- Judge colour only in bright, standardised light: rods give no colour, and adaptation and contrast distort casual viewing.
- Screen shade passers (Ishihara, Farnsworth–Munsell 100 Hue); about 8% of men have a red–green defect.
- D65/10° is the textile standard because real samples fill a large visual field.

## 11. Summary

The eye focuses light onto a retina where rods serve night and cones serve colour. Three cone classes make colour a three-number signal; opponent channels make it a perceptual one; adaptation and contrast make it a contextual one. Those strengths and weaknesses explain the whole apparatus of colour control — standard illuminants, neutral cabinets, screened observers, standard-observer tables and instrumental arbitration.

## 12. Practice

1. Explain why a red and a black fabric become hard to distinguish at dusk, naming the receptors and the shift involved.
2. A new shade passer confuses red and green samples but sees blue–yellow differences normally. Name the two most likely deficiencies and one screening test.
3. Why does CIELAB use red–green and yellow–blue axes rather than red, green and blue axes?
4. A buyer views a large fabric panel; a laboratory technician views a 1 cm aperture. Which standard observer fits each, and why is 10° the textile default?
5. State two viewing rules that control simultaneous contrast and adaptation in a light cabinet.

### Answer key

1. At dusk vision shifts from cones (photopic) to rods (scotopic) — the Purkinje shift: peak sensitivity moves from 555 nm to ≈ 507 nm, reds darken strongly and rods report only grey levels, so red and black converge.
2. Protanopia (L cones absent) or deuteranopia (M cones absent) — both give red–green confusion; screen with Ishihara plates (and grade discrimination with the Farnsworth–Munsell 100 Hue test).
3. Because perception is opponent-organised after the cone stage: red opposes green and yellow opposes blue, so those axes match how differences are actually seen (and no reddish-green exists).

4. Panel → 10° observer (large field, fovea plus surrounding retina); small aperture → 2° observer. Textile samples are viewed large, so D65/10° is the default specification.
5. Use the neutral grey cabinet/surround with samples touching and no gaps, and allow adaptation time under each illuminant before judging (any two of: fixed geometry, no outside light, standard illumination level).

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